

Normal and Binomial Distributions

Models for continuous outcomes and repeated yes/no trials

Sergio I. García-Ríos

PA 311N: Data Analysis for Public Policy

LBJ School of Public Affairs

September 28, 2026

From single events to distributions

Last time we asked questions like:

$$P(A), \quad P(A \mid B), \quad P(A \cap B).$$

Today we ask a different kind of question:

What values can a **random variable** take,
and how likely is each value?

A probability distribution gives us a model for the full range of possible outcomes.

Today: two models for uncertainty

1. **Probability distributions** organize possible values and their probabilities.
2. The **normal distribution** models many continuous, symmetric outcomes.
3. **Z-scores** put observations on a common standard-deviation scale.
4. The **binomial distribution** models counts of successes in repeated binary trials.

We will move back and forth between **pictures**, **formulas**, and **R**.

A random variable turns outcomes into numbers

A **random variable** assigns a numerical value to the outcome of a random process.

Suppose we randomly select 4 households eligible for a weatherization program.

- ▶ Each household either completes enrollment or does not.
- ▶ Let X = number of households that complete enrollment.

Then

$$X \in \{0, 1, 2, 3, 4\}.$$

The uncertainty is no longer just “success or failure.” It is about the **value of X** .

A probability distribution describes the random variable

A probability distribution tells us:

1. which values the random variable can take; and
2. the probability attached to those values or ranges of values.

Every probability distribution must obey:

$$0 \leq P(\text{event}) \leq 1$$

and the probabilities across all possible outcomes must total 1.

Different kinds of random variables require different kinds of distributions.

Discrete and continuous distributions

Discrete

- ▶ Separate, countable values
- ▶ We can assign probability to an exact value
- ▶ Example: number of households enrolled
- ▶ Binomial distribution

Continuous

- ▶ Values along a continuum
- ▶ Probability comes from intervals, not exact points
- ▶ Example: height or a test score model
- ▶ Normal distribution

Discrete: probability at values. Continuous: probability over ranges.

Your turn: discrete or continuous?

Your Turn

Classify each random variable.

1. Number of 311 calls received in one hour
2. Time required to process a permit application
3. Number of surveyed residents who support a proposal
4. Household annual income

Your turn: discrete or continuous?

Your Turn

Classify each random variable.

1. Number of 311 calls received in one hour
2. Time required to process a permit application
3. Number of surveyed residents who support a proposal
4. Household annual income

Discrete, continuous, discrete, continuous.

The normal distribution is a family of curves

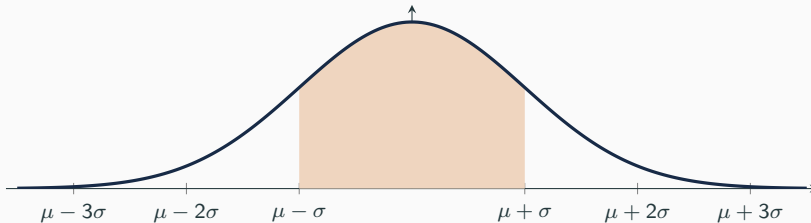
A normal distribution is characterized by two parameters:

$$X \sim N(\mu, \sigma)$$

- ▶ μ determines the center.
- ▶ σ determines the spread.
- ▶ The curve is symmetric and unimodal.
- ▶ Total area under the curve equals 1.

Changing μ moves the curve. Changing σ stretches or compresses it.

The 68–95–99.7 rule



Within 1σ	$\approx 68\%$	of observations
Within 2σ	$\approx 95\%$	of observations
Within 3σ	$\approx 99.7\%$	of observations

Your turn: use the empirical rule

Suppose scores on a large assessment are approximately normal with

$$\mu = 70, \quad \sigma = 10.$$

Your Turn

Without a calculator, approximately what proportion of scores fall:

1. between 60 and 80?
2. between 50 and 90?
3. above 80?

Your turn: use the empirical rule

Suppose scores on a large assessment are approximately normal with

$$\mu = 70, \quad \sigma = 10.$$

Your Turn

Without a calculator, approximately what proportion of scores fall:

1. between 60 and 80?
2. between 50 and 90?
3. above 80?

, , .

Z-scores put observations on a common scale

A z-score tells us how many standard deviations an observation falls above or below the mean:

$$z = \frac{x - \mu}{\sigma}$$

- ▶ $z = 0$: exactly at the mean
- ▶ $z = 1$: one standard deviation above the mean
- ▶ $z = -2$: two standard deviations below the mean

Z-scores remove the original units and replace them with **standard-deviation units**.

Our turn: standardize an observation

Suppose assessment scores have

$$\mu = 70, \quad \sigma = 10.$$

A student scores 88.

$$z = \frac{88 - 70}{10} = \boxed{1.8}$$

Our Turn

Interpret $z = 1.8$ in words.

Our turn: standardize an observation

Suppose assessment scores have

$$\mu = 70, \quad \sigma = 10.$$

A student scores 88.

$$z = \frac{88 - 70}{10} = \boxed{1.8}$$

Our Turn

Interpret $z = 1.8$ in words.

The score is 1.8 standard deviations above the mean.

Z-scores let us compare different scales

Consider two observations:

Assessment A

$$\mu = 70, \sigma = 10, x = 90$$

$$z = 2$$

Assessment B

$$\mu = 500, \sigma = 50, x = 600$$

$$z = 2$$

The raw scores are not comparable, but both observations occupy the same relative position in their distributions.

Important distinction: z-scores vs. normal probabilities

You can calculate a z-score for **any** quantitative distribution with a mean and standard deviation.

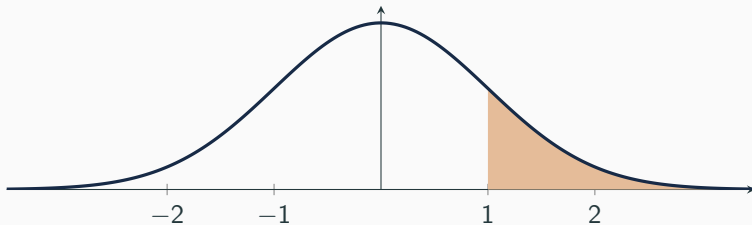
But translating a z-score into a normal probability requires an approximately normal model.

Do not skip the shape

A z-score of 2 always means “two SDs above the mean.” It does **not** always mean that only about 2.5% of observations are above it.

This is why we looked at distributions before calculating summaries earlier in the course.

For a continuous distribution, probability is area



$$P(Z > 1) \approx 0.159.$$

The probability is the **area under the curve** over the region we care about.

R automates normal probabilities

For $X \sim N(70, 10)$, what is the probability that $X > 80$?

R

```
1 - pnorm(80, mean = 70, sd = 10)
```

$$P(X > 80) \approx \boxed{0.1587}.$$

- ▶ `pnorm()` gives area to the left.
- ▶ For an upper tail, use $1 -$ that area.

Normal is a model, not a default

Some variables are approximately symmetric and bell-shaped. Others are not.

Normal model may be reasonable

- ▶ adult height in a homogeneous population
- ▶ some standardized measurement errors
- ▶ some large-scale assessment scores

Often not normal

- ▶ household income
- ▶ emergency-room wait time
- ▶ counts of events
- ▶ heavily bounded or skewed outcomes

Look at the variable before choosing the model.

Repeated yes/no trials lead to a different model.

The binomial random variable counts successes

Suppose a city outreach program has a 60% enrollment-completion rate among eligible households.

We randomly select 4 households and define

X = number that complete enrollment.

Possible values are

$$X \in \{0, 1, 2, 3, 4\}.$$

The individual outcome is binary. The **binomial random variable** counts how many successes occur across repeated trials.

When can we use a binomial model? Think BINS

A binomial model requires:

B Binary outcome on each trial: success/failure.

I Independent trials.

N Fixed number of trials, n .

S Same probability of success, p , on each trial.

If these conditions hold, then

$$X \sim \text{Binom}(n, p).$$

Your turn: is a binomial model appropriate?

Your Turn

A city randomly samples 20 permit applications and counts how many are approved within 10 business days. Can we model the count as binomial? What would you need to assume?

Your turn: is a binomial model appropriate?

Your Turn

A city randomly samples 20 permit applications and counts how many are approved within 10 business days. Can we model the count as binomial? What would you need to assume?

We need a binary outcome, fixed $n = 20$, approximately independent applications, and a common probability p of timely approval.

The formula is easy. The model assumptions require judgment.

Exactly two successes in four trials

Suppose $p = 0.60$ and $n = 4$. What is

$$P(X = 2)?$$

One possible sequence is

$$S, S, F, F.$$

Its probability is

$$(0.60)^2(0.40)^2 = 0.0576.$$

But there are several different sequences with exactly two successes.

How many sequences produce exactly k successes?

With $n = 4$ and $k = 2$, the possible success positions are:

SSFF, SFSF, SFFS, FSSF, FSFS, FFSS.

There are 6 sequences.

In general, the number of ways to choose k success positions from n trials is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

For $n = 4$, $k = 2$:

$$\binom{4}{2} = 6.$$

The binomial probability formula

For $X \sim \text{Binom}(n, p)$,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Think of the formula as two pieces:

1. Probability of one particular sequence: $p^k (1 - p)^{n-k}$
2. Number of sequences: $\binom{n}{k}$

$$P(X = k) = \text{number of ways} \times \text{probability of each way.}$$

Our turn: exactly two of four

For $n = 4$, $p = 0.60$, and $k = 2$:

$$\begin{aligned}P(X = 2) &= \binom{4}{2} (0.60)^2 (0.40)^2 \\&= 6(0.36)(0.16) \\&= \boxed{0.3456}.\end{aligned}$$

Our Turn

Interpret 0.3456 in words.

Our turn: exactly two of four

For $n = 4$, $p = 0.60$, and $k = 2$:

$$\begin{aligned}P(X = 2) &= \binom{4}{2} (0.60)^2 (0.40)^2 \\&= 6(0.36)(0.16) \\&= \boxed{0.3456}.\end{aligned}$$

Our Turn

Interpret 0.3456 in words.

About 34.6% of repeated groups of four would contain exactly two completions under this model.

Your turn: exactly four of five

Suppose $X \sim \text{Binom}(5, 0.60)$.

Your Turn

Calculate $P(X = 4)$ by hand. Identify n , k , p , and $1 - p$ before substituting.

Your turn: exactly four of five

Suppose $X \sim \text{Binom}(5, 0.60)$.

Your Turn

Calculate $P(X = 4)$ by hand. Identify n , k , p , and $1 - p$ before substituting.

$$\begin{aligned}P(X = 4) &= \binom{5}{4} (0.60)^4 (0.40)^1 \\&= 5(0.1296)(0.40) \\&= \boxed{0.2592}.\end{aligned}$$

R automates exact binomial probabilities

The R function is `dbinom()`:

R

```
dbinom(4, size = 5, prob = 0.60)
```

$$P(X = 4) = 0.2592.$$

The arguments map directly onto the model:

- ▶ first argument: k , desired number of successes
- ▶ size: n , number of trials
- ▶ prob: p , probability of success

Exactly, at most, and at least are different questions

For $X \sim \text{Binom}(20, 0.60)$:

$P(X = 15)$ exactly 15

$P(X \leq 15)$ at most 15

$P(X \geq 15)$ at least 15

R

```
dbinom(15, 20, .60)      # exactly 15
pbinom(15, 20, .60)      # at most 15
1 - pbinom(14, 20, .60)  # at least 15
```

A binomial distribution has a center and spread

If

$$X \sim \text{Binom}(n, p),$$

then

$$\mu = np$$

and

$$\sigma = \sqrt{np(1-p)}.$$

These formulas describe the distribution of the **count of successes** across repeated samples.

Our turn: expected completions in 20 households

Suppose

$$X \sim \text{Binom}(20, 0.60).$$

Expected number of completions:

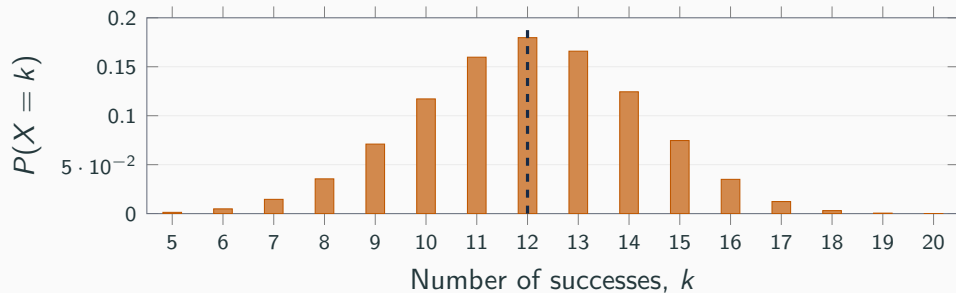
$$\mu = np = 20(0.60) = \boxed{12}.$$

Standard deviation:

$$\sigma = \sqrt{20(0.60)(0.40)} \approx \boxed{2.19}.$$

This does **not** mean every group of 20 contains exactly 12 successes. It describes the center and typical variation across repeated groups.

What does the binomial distribution look like?



For $n = 20, p = .60$, the distribution is centered at $np = 12$.

Why normal and binomial appear in the same lecture

As n grows, many binomial distributions become increasingly smooth and bell-shaped.

A common rule of thumb checks the expected number of successes and failures:

$$np \geq 10 \quad \text{and} \quad n(1 - p) \geq 10.$$

For $n = 20, p = .60$:

$$np = 12, \quad n(1 - p) = 8.$$

So this example is **not quite** in the comfortable range for a normal approximation.

Two distributions, two kinds of questions

	Normal	Binomial
Variable	continuous	count of successes
Parameters	μ, σ	n, p
Probability	area over intervals	probability at integer counts
R functions	<code>pnorm()</code>	<code>dbinom()</code> , <code>pbinom()</code>
Key check	shape is plausible	BINS conditions hold

Your Turn

For each statement, identify the model or calculation you would use.

1. A score is 1.5 standard deviations above its mean.
2. Exactly 7 of 10 independent respondents support a proposal, with $p = .60$.
3. A variable is strongly right-skewed. Should you automatically use a normal model?
4. In a binomial model with $n = 50$, $p = .30$, what is the expected number of successes?

Your Turn

For each statement, identify the model or calculation you would use.

1. A score is 1.5 standard deviations above its mean.
2. Exactly 7 of 10 independent respondents support a proposal, with $p = .60$.
3. A variable is strongly right-skewed. Should you automatically use a normal model?
4. In a binomial model with $n = 50$, $p = .30$, what is the expected number of successes?

Z-score; binomial; no; $np = 15$.

What I want you to leave with

1. A **distribution** describes the possible values of a random variable and how probability is allocated across them.
2. The **normal model** is continuous, symmetric, and summarized by μ and σ .
3. A **z-score** measures distance from the mean in standard-deviation units.
4. The **binomial model** counts successes when BINS conditions hold.
5. The formulas matter, but choosing the right model comes first.

Identify the variable. Check the model. Calculate. Interpret.